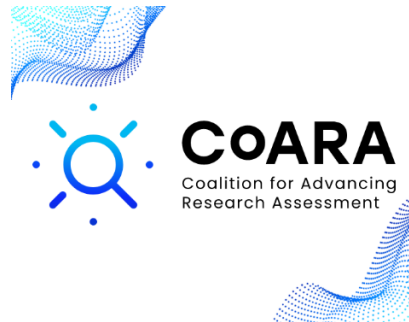




Action plan 2023-2027 for advancing research assessment at ISTA

Introduction

The [Institute of Science and Technology Austria \(ISTA\)](#) is an institute for basic research in science and science-related technology areas with the right to grant Ph.D. degrees (with a M.Sc. option on the way to a Ph.D.). The Institute was founded through an Austrian federal law in 2006. The campus in Klosterneuburg near Vienna, opened in 2009. The Institute is independent of other institutions and organizations in Austria. With its foundation, it was given a global 10-year budget that allowed the Institute to grow to up to 45 research groups by 2017. Following recommendations of external evaluations of the Institute, the Austrian government granted budgets which will allow the Institute to grow to up to 150 research groups by 2036.

Within this framework, and with the mission to perform research and doctoral education on a world-class level, the Institute has had complete freedom in building a campus for science, hiring scientists, and creating the internal structures and the culture of the Institute. Currently, our scientists perform theoretical and experimental research across a broad range of the sciences, without any departmental subdivision. The research groups are organized in small to medium-size research groups that interact with each other across the boundaries of traditional disciplines. Each research group of ISTA is headed by a tenured or tenure-track professor.

The structures of ISTA have been set up to allow the Institute to compete with the top institutions in the world for professors, Ph.D. students, and postdocs. These structures include a tenure track for professors and an English-language graduate school that recruits internationally. Professors are not recruited in predefined fields, but mainly according to their research potential. The culture at the Institute emphasizes curiosity-driven research by interacting, small, independent groups that consist of group leaders who supervise Ph.D. students and postdocs. In support of this culture, research groups share office and laboratory spaces, as well as scientific equipment and services wherever feasible. All group leaders receive core funding from the Institute, but are encouraged and incentivized to also acquire external funds for their research.

Context and applications

ISTA recognizes the need to evaluate the way it assesses research, aiming to broaden the diversity of recognized research outputs and activities, whilst maintaining the quality and impact of research. ISTA has already developed a set of rules and procedures for various aspects of research assessment. These rules have to be developed further along the principles mentioned above.

In November 2022, ISTA signed two declarations: DORA (Declaration on Research Assessment), and the Agreement on Reforming Research Assessment from CoARA (Coalition for Advancing Research Assessment). ISTA takes these actions as the start of a journey and an opportunity for institutional learning. The Scientific Board, which is in charge of overseeing the assessment of scientific quality, recommended that ISTA sign DORA and overall supports the goals of the CoARA Agreement.

ISTA has identified the following ISTA-specific processes which will need to be re-evaluated or revised in light of the CoARA principles:

1. Faculty hiring process
2. Tenure evaluations of Assistant Professors
3. Hiring of PhD Students, Postdocs and Staff Scientists
4. Career development programs

5. Topical Reviews/Scientific evaluations
6. Progress reviews of PhD Students and Postdocs

How ISTA has started the process

The Institute management decided that the first process to be evaluated was faculty hiring. Later, it shall widen its scope to tenure evaluations, assessments of other scientific positions (staff scientists, postdocs, students) and departments or the whole institute. With these principles in mind, a working group was established.

The Research Assessment Reform Working Group (RARwg) was established in December 2022.

Its current members (status October 2024):

Prof. Nick Barton (Evolutionary Biology; Chair of the working group, Chair of the Institute Committee)

Prof. Robert Seiringer (Mathematical Physics) (since September 2024) replaced Prof. Jan Maas

(Mathematics; MPS Recruiting Committee member)

Prof. Tim Vogels (Theoretical Neuroscience; Professorial Committee member)

Prof. Daniel Zilberman (Plant Science; LS Recruiting Committee member)

Hilde Janssens, PhD (Human Resources, EDI Officer)

Catherine Lloyd, PhD (Faculty Office) (since May 2024)

Snezana Vuletic Solari, PhD (Office of the President, Academic Appointments & Strategy Support) (since

May 2024) replaced Laurenz Niel, PhD (Office of the President, Professorial Appointments and

Institutional Research)

Helga Materna, PhD (Faculty Recruiting and Evaluations) (since January 2024) replaced Irma Tristan, PhD

(Faculty Recruiting and Evaluations)

Action plan

The action plan is a living document and will be updated regularly.

Specific rules and procedures for research assessment at ISTA must be approved by different groups:

- Board of Trustees: High-level Rules for Professors / Employees
- Executive Committee: Rules for Tenure Evaluations
- Scientific Board: General Oversight on quality assurance, Rules for Institute Evaluations, topical reviews, research group evaluations, mid-term evaluations of Assistant Professors, progress reviews of postdocs etc.
- Advisory Committee of the Graduate School: quality assessments of students (progress reviews, qualifying exams, thesis defenses)

The chair of the RARwg will discuss progress, recommendations and plans with the Scientific Board at its annual meetings in June.

Objectives

Short-term outlook: sharing with the CoARA community an action plan by the end of 2024, participating in the current discussions of faculty recruitment procedures, which aim for concrete changes (assessment criteria, tools) for this and next hiring season.

Long-term outlook: developing and evaluating criteria, tools and processes that fulfill the core Commitments for CoARA, , by 2027 we should have worked through at least one cycle of review and development of our assessment criteria, tools and processes.

Actors

RAR working group -Area Chairs -Faculty -President -Faculty recruiting and Evaluations team –Academic and Scientific Staff that participates in recruiting and scientific evaluations -Ombudspersons

The actions and initiatives came specifically from the RAR working group.

1) Faculty recruiting

ISTA recruits new faculty via an open call which usually covers the wide range of current fields at the institution, and in occasion, extends to new ones. This presents both challenges and opportunities when it comes to evaluating and comparing candidates across these fields, which have led us to years of active reflection on research assessment that goes beyond the usual quantitative markers. We believe that ISTA already does some things well, such as not heavily relying on quantitative indicators and conducting extensive interviews, which contribute to a holistic view of candidates. As faculty recruitment is a central process at our institution, it also sets the tone for other evaluations such as tenure and research group evaluations.

ISTA is thus committed to the main goals of research assessment advancement: quality over quantity; broader set of criteria beyond classical journal publications; transparency. In summer 2022, the Institute re-defined its hiring criteria, listed in Appendix A. ISTA recognizes that it can further improve its practices.

Following the principles of DORA and CoARA will also foster the right environment to achieve greater diversity at ISTA. Our current concrete goals and measures in particular to Equity & Inclusion are summarized in our [Gender Equality Plan](#).

- **Structured interview revision**

Candidates that are interviewed during our faculty recruiting process, are invited to a two-day visit with a dense program in order to get a holistic view of the candidate and their application. The recruiting committees produced a set of questions that would be asked in the Q&A, a question and answer session which is a part of the interview. This will help keep the process fair while evaluating the candidates.

- **Community Engagement Statement (CES) revision and implementation**

Shortlisted candidates will be asked to provide a Community Engagement Statement of no more than 2 pages addressing the questions:

What are the pressing challenges you see for your scientific community?

How have you addressed these challenges in the past and how would you like to address them in the future?

Applicants are asked to provide a narrative statement - not simply a list of committees or activities – regarding initiatives they have been/are/would like to be involved in. Topics may include: leadership, building a positive research culture, diversity & inclusion, research ethics & integrity, open access, sustainability, science education, outreach, contributions to policy, tech transfer etc. Applicants should explain why they chose these topics, their efforts in making a positive contribution, and their outcomes. This statement will be a basis for discussion as part of the interview.

This measure was applied with the faculty candidate interviews starting in winter 2023.

We will critically evaluate the pros and cons of the CES during and after the hiring season to improve or correct the measure in summer 2024 and 2025.

- **Observer role at faculty recruiting discussions**

The faculty hiring process at ISTA includes the participation of an “observer” during faculty candidate discussions. The observer facilitates communication between the committees, and also encourages and reinforces good practices. From 2017 till 2022, this task was executed by a member of the administrative staff, i.e. the EDI officer. This year this role was assigned to the RC member that partly overlaps with the RAR working group, in the hope that placing this responsibility with the faculty itself will result in more dedication and commitment to fair evaluations.

This measure started with the first faculty candidate discussion on fall 2023.

We will evaluate and revise the measure after two recruiting seasons in summer 2025.

- **Other possible courses of action**

Narrative CV study and possible implementation: ISTA will study the possibility of requesting a narrative CV from applicants to faculty positions where applicants would emphasise a few key achievements following the example of other institutions, for example the Royal Society, EMBL, NWO or SNF. In case the narrative CV will be implemented in recruiting, it could be also considered for tenure evaluations. Discussion is still ongoing.

2) Revision of the tenure evaluation process at ISTA

The working group has discussed the current tenure evaluation process at ISTA and as consequence will develop a plan for recommended measures to be taken to improve the process in accordance with responsible research assessment principles. In general, evaluations should aim to balance academic achievements with contributions to the scientific community and future potential.

- **Redefine criteria and include indicators**

The RAR working group discussed and recommended updated criteria for the new rules for Tenure Evaluations at ISTA in summer 2024. Further discussions and recommendations may lead to define indicators. The definition of indicators should help to provide a more transparent communication, more clarity and an improved guidance through the process.

3) Revision of Scientific Evaluations/Topical Reviews

ISTA will reassess its area-wide/field-specific evaluation processes to align with responsible research assessment practices. A revised process should result in more relevant evaluations and higher stakeholder satisfaction. Evaluation and reflection on the current processes should start end of 2024 and recommendations will be tackled in 2025. Detailed actions to be determined.

4) Workshops with other committees/offices with compatible goals

ISTA will raise awareness on the research assessment reform and the RARwg will bundle its current efforts with other committees/groups to coordinate and bring together resources and ideas. Relevant Committees/Offices include: the Research Ethics Office; the Equity, Diversity & Inclusion Working Group; Faculty Office; Postdoctoral Office.

Collaborations with committees/groups that are working on aspects of research assessment at ISTA should start in 2025 (organisation of workshops in summer 2025).

5) Hiring of PhD students, Postdocs and Staff Scientists

ISTA will evaluate its recruitment processes to ensure that they reflect diverse research contributions. A more diverse pool of applicants will also enhance recruitment satisfaction and should bring equitable hiring practices across all research positions at ISTA.

Evaluation and reflection on the current processes should start in fall 2025 and recommendations will be tackled in 2026 (in parallel with action 5 and 6).

6) Career development programs

ISTA will enhance career development to support diverse trajectories in research careers. The evaluation of the current program and suggestions for new initiatives should build up a greater career satisfaction and eventually also lead to the retention of talent.

Evaluation and reflection on the current processes should start in fall 2025 and recommendations will be tackled in 2026 (in parallel with action 4 and 6). Detailed actions to be determined.

7) Reflection on progress reviews of PhD students and Postdocs

The current processes should be reviewed to see whether they recognise a broader spectrum of contributions. Fair evaluations should foster diverse career pathways.

Evaluation and reflection on the current processes should start fall 2025 and recommendations will be tackled in 2026 (in parallel with action 4 and 5). Detailed actions to be determined.

By implementing this action plan, ISTA aims to enhance its research assessment framework in line with international best practices, fostering a more inclusive and dynamic academic community.

Appendices

Appendix A: Current Evaluation Criteria of ISTA in faculty recruitment

Past scientific achievements

- Impact/novelty of research
- Publication record (emphasize quality) (incl data sharing, open access publications, software)
- Acquired research funding
- Awards and distinctions

Future scientific potential

- Vision (short and long term)
- Originality of research topics & methodology (incl. independence)
- Potential for impact in the field and more broadly
- Fit of research fields to ISTA

Teaching, contributions, diversity

- Teaching and communication skills
- Service to the scientific community
- Inter-sectoral activity/visibility
- Demonstrate ability to mentor and supervise
- Leadership potential in the field and at the Institute
- Contributions to diversity & inclusion